SYED NAZIYA PARVEEN

2303A52394

BATCH-42

Task 1: Password Strength & Validation Check

Prompt:

Generate a simple Python user registration on program and then improve it by enforcing strong password validation including minimum length of at least 8 characters, requiring a mix of uppercase, lowercase, digits and special characters using regular expressions, and rejecting common or weak passwords, finally output the secure version of the script with proper validation implemented

Code:

```
#Generate a simple Python user registration program and then improve it by enforcing strong password validation including minimum length of at least 8 characters, requiring a mix of uppercase, lowercase, digits and special characters using regular expressions, and rejecting common or weak passwords, finally output the secure version of the script with proper validation implemented
# Simple User Registration Program
'''def register_user():
    username = input("Enter a username: ")     password = input("Enter a
password: ")     print(f"User '{username}' registered successfully!")
register_user()'''
# Improved User Registration Program with Strong Password Validation import re def
is_strong_password(password):
    if len(password) < 8:
        return False     if not re.search(r'[A-Z]', password):
        return False     if not re.search(r'[a-z]', password):
        return False     if not re.search(r'[0-9]', password):
        return False     if not re.search(r'[@$!%*?&]',
password):
        return False
    common_passwords = ['password', '123456', '12345678', 'qwerty', 'abc123']        if password.lower() in
common_passwords:
        return False     return
True def register_user():
    username = input("Enter a username: ")     while True:
        password = input("Enter a password: ")         if
is_strong_password(password):
            print(f"User '{username}' registered successfully!")            break         else:
            print("Weak password. Please enter a stronger password that is at least 8 characters long, includes uppercase and
lowercase letters, digits, and special characters, and is not a common password.")
register_user()
```

Output:

```
Enter a username: naziya
Enter a password: naziya@@
Weak password. Please enter a stronger password that is at least 8 characters long, includes uppercase and lowercase letters, digits, and special characters, and is not a common password.
Enter a password: Syednaziya@@
User 'naziya' registered successfully!
```

JUSTIFICATION:

This code creates a user registration system that checks if the entered password is strong and secure before accep ng it. It prevents weak passwords by enforcing rules like minimum length, character variety, and blocking commonly used passwords.

Task 2: Insecure File Upload Handling

Prompt:

Generate a Python Flask file upload program and improve it by adding security features such as validating allowed file extensions, restricting file size limits, and safely storing uploaded files using secure paths and filenames to prevent malicious uploads

Code:

```
from flask import Flask, request, render_template_string
import os from werkzeug.utils import secure_filename

app = Flask(__name__)

# Configuration
UPLOAD_FOLDER = 'uploads'
MAX_FILE_SIZE = 16 * 1024 * 1024  # 16 MB
ALLOWED_EXTENSIONS = {'txt', 'pdf', 'png', 'jpg', 'jpeg', 'gif'}
 app.config['UPLOAD_FOLDER'] =
UPLOAD_FOLDER
app.config['MAX_CONTENT_LENGTH'] =
MAX_FILE_SIZE

# Ensure upload folder exists if not
os.path.exists(UPLOAD_FOLDER):
    os.makedirs(UPLOAD_FOLDER)

# Check allowed file types
def allowed_file(filename):
    return '.' in filename and filename.rsplit('.', 1)[1].lower() in ALLOWED_EXTENSIONS

# Simple HTML form
HTML = '''
<!doctype html>
<title>Upload File</title>
<h2>Upload File</h2>
<form method="post" enctype="multipart/form-data">
```

```
<input type="file" name="file">
```

```
  <input type="submit" value="Upload">
</form>
<p>{{ message }}</p>
'''

@app.route('/', methods=['GET',
'POST']) def upload_file():     message =
'''
    if request.method ==
'POST':
        if 'file' not in request.files:
message = "No file part"        else:
            file = request.files['file']
             if file.filename == '':
                message = "No selected file"
             elif not allowed_file(file.filename):
                message = "Invalid file type"

else:
                filename = secure_filename(file.filename)              file_path =
os.path.join(app.config['UPLOAD_FOLDER'], filename)              file.save(file_path)
message = "File uploaded securely"

    return render_template_string(HTML, message=message)
 if __name__ ==
'__main__':
app.run(debug=True)
```

Output:

JUSTIFICATION:

This code builds a file upload system and checks for security risks before accepting any file. It prevents harmful uploads by restricting file types, limiting file size, and storing files safely.

Task 3: Real-Time Applica on: API Security Review

Prompt:

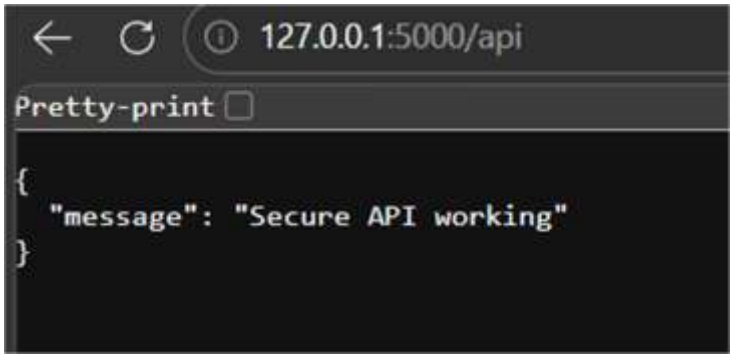
Generate a Python Flask API endpoint and then review and improve it by identifying security issues like missing authentication, unsafe HTTP methods, and no rate limiting, and fix them by adding token-based authentication, proper input validation, and rate limiting to create a secure API

Code:

```
from flask import Flask, request, jsonify from functools import wraps
import time
 app = Flask(__name__)

# In-memory token store (demo only) tokens = {"valid_token": "user1"}

# Token authentication def token_required(f):
    @wraps(f)     def decorated(*args, **kwargs):
        token = request.headers.get('Authorization')         if not token or token not in tokens:
            return jsonify({"message": "Token is missing or invalid!"}), 401         return f(*args, **kwargs)     return
decorated

# Simple rate limiter rate_limit = {}
 def rate_limiter(f):     @wraps(f)     def decorated(*args,
**kwargs):
        token = request.headers.get('Authorization')         user = tokens.get(token)
         if user:
            current_time = time.time()             if user in rate_limit and current_time - rate_limit[user] < 60:
return jsonify({"message": "Rate limit exceeded!"}), 429             rate_limit[user] = current_time
         return f(*args, **kwargs)     return decorated

# Home route (fixes your 404 issue)
@app.route('/') def home():
    return "API is running"

# Test API route
@app.route('/api', methods=['GET']) def api():
```

```
    return jsonify({"message": "Secure API working"})

# Secure endpoint
@app.route('/data', methods=['POST'])
@token_required
@rate_limiter def
data():
    data = request.get_json()
     if not data or 'value' not in data:
        return jsonify({"message": "Invalid input!"}), 400
     return jsonify({
        "message": "Data received!",
        "data": data['value']
    }), 200
 if __name__ == '__main__':
app.run(debug=True)
```

Output:



JUSTIFICATION:

This code reviews an API endpoint to find security weaknesses like lack of authentication on and validation. It then improves security by adding token-based access control, validating inputs, and limiting request rates to prevent misuse.

Task 4: Cross-Site Scrip ng (XSS) Check

Prompt:

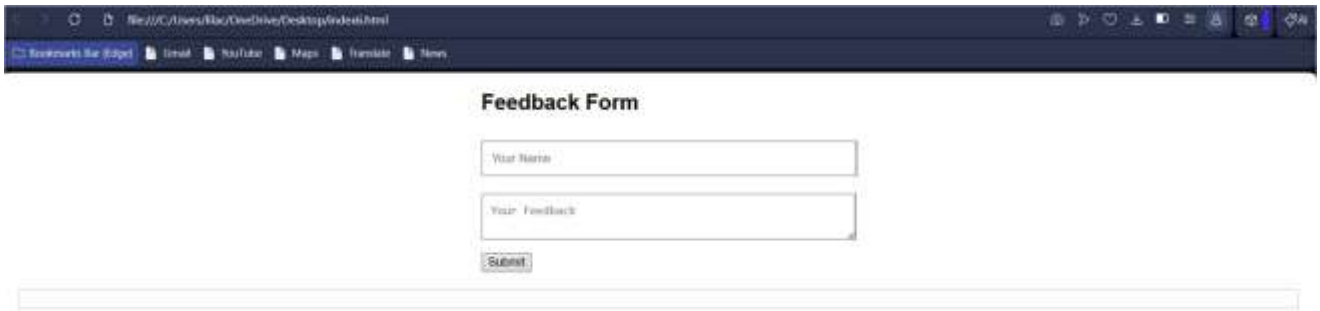
Generate an HTML feedback form with JavaScript that displays user input on the page and then improve it by identifying XSS vulnerabilities such as directly rendering untrusted input, and fix them by escaping HTML entities and applying security measures like Content Security Policy (CSP) to prevent script injection

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Feedback Form</title>
   <style>         body {             font-family: Arial, sans-serif;             margin:
20px;
       }
       .feedback-form {            max-width: 400px;            margin:
auto;
       }
       .feedback-form input, .feedback-form textarea {            width: 100%;            padding:
10px;            margin: 10px 0;         }
       .feedback-display {            margin-top: 20px;            padding:
10px;
           border: 1px solid #ccc;
       }
   </style>
</head>
<body>
   <div class="feedback-form">
       <h2>Feedback Form</h2>
       <input type="text" id="name" placeholder="Your Name">
       <textarea id="feedback" placeholder="Your Feedback"></textarea>
       <button onclick="submitFeedback()">Submit</button>
   </div>
   <div class="feedback-display" id="feedbackDisplay"></div>
   <script>
       function escapeHtml(unsafe) {            return unsafe
               .replace(/&/g, "&amp;")
               .replace(/</g, "&lt;")
               .replace(/>/g, "&gt;")
               .replace(/"/g, "&quot;")
               .replace(/'/g, "&#039;");
       }
        function submitFeedback() {            const name = document.getElementById('name').value;
           const feedback = document.getElementById('feedback').value;
           // Escape user input to prevent XSS            const safeName = escapeHtml(name);
```

```
        const safeFeedback = escapeHtml(feedback);

        // Display the feedback safely          const feedbackDisplay =
document.getElementById('feedbackDisplay');
        feedbackDisplay.innerHTML = `<strong>${safeName}:</strong> ${safeFeedback}`;
    }
  </script>

  <!-- Content Security Policy to prevent script injection -->
  <meta http-equiv="Content-Security-Policy" content="default-src 'self'; script-src
'self'; style-src 'self';">
</body>
</html>
```

Output:



Justification

This code checks if user input is safely handled before displaying it in the browser. It prevents XSS a acks by escaping harmful scripts and enforcing security policies like CSP.

Task 5: SQL Injection Prevention

Prompt:

Generate a Python script that uses SQLite to fetch user details from a database and then improve it by identifying SQL injection vulnerabilities such as string concatenation in queries and fixing them using parameterized queries (prepared statements) to make it secure

Code:

```
import sqlite3

# Create and initialize database (only runs once safely) def setup_database():
  conn = sqlite3.connect('users.db')     cursor =
conn.cursor()

  cursor.execute('''
      CREATE TABLE IF NOT EXISTS users (
id INTEGER PRIMARY KEY,          name TEXT
      )
  ''')
```

```
    # Insert sample data (only if empty)     cursor.execute("SELECT COUNT(*) FROM
users")     if cursor.fetchone()[0] == 0:
        cursor.execute("INSERT INTO users (name) VALUES ('alice')")        cursor.execute("INSERT INTO users (name)
VALUES ('bob')")        cursor.execute("INSERT INTO users (name) VALUES ('likitha')")        conn.commit()
    conn.close()

# Insecure function (SQL Injection vulnerable) def get_user_insecure(user_id):
    conn = sqlite3.connect('users.db')     cursor = conn.cursor()
    query = f"SELECT * FROM users WHERE id = {user_id}"  # ❌ vulnerable     cursor.execute(query)     result =
cursor.fetchall()
    conn.close()     return result

# Secure function (using parameterized query) def get_user_secure(user_id):
    conn = sqlite3.connect('users.db')     cursor = conn.cursor()

    query = "SELECT * FROM users WHERE id = ?"  # ☑ safe     cursor.execute(query, (user_id,))
result = cursor.fetchall()
    conn.close()     return result

# Main program if __name__ == "__main__":
    setup_database()
    user_id = input("Enter user ID: ")

    print("\nFetching user details (insecure):")     try:
        print(get_user_insecure(user_id))     except Exception as e:
print("Error:", e)

    print("\nFetching user details (secure):")     try:
        print(get_user_secure(user_id))     except Exception as e:
print("Error:", e)
```

Output:

```
erver/ai5.py"
Enter user ID: 1

Fetching user details (insecure):
[(1, 'alice')]

Fetching user details (secure):
[(1, 'alice')]
```

JUSTIFICATION:

This code checks how user input is used in database queries and iden fies risks of SQL injec on. It prevents a acks by using parameterized queries instead of directly inser ng user input into SQL statements.